

Predicting Employee Attrition Using Machine Learning for Human Resource Decision Support

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Business Problem

Employee attrition is a costly organizational problem because voluntary departures can increase recruiting, onboarding, training, and productivity-recovery costs while also reducing institutional knowledge and team continuity. The purpose of this project is to move beyond descriptive turnover reporting and evaluate whether employee characteristics and workplace conditions can be used to identify patterns associated with attrition. The analysis is designed as a decision support tool for Human Resources (HR), not as an automated system for deciding who should be hired, promoted, disciplined, or terminated. The central business questions are which factors are most strongly associated with attrition, whether overtime materially changes attrition risk, how compensation and satisfaction relate to retention, and which classification model provides the most useful balance of predictive performance and interpretability.

The business value of addressing employee attrition extends beyond reducing the direct cost of replacing employees. Frequent turnover can disrupt teams, increase workloads for remaining employees, reduce productivity, and create challenges in maintaining organizational knowledge. An organization that can recognize patterns associated with attrition may be able to intervene at the organizational level before these problems become more severe. For this reason, the purpose of the analysis is not simply to predict whether an employee might leave, but to identify workplace conditions that management can evaluate and potentially improve. The project therefore connects machine learning performance with a practical HR objective: using data to support better retention strategies while maintaining appropriate human oversight.

Background/History

Organizations have traditionally monitored turnover after employees leave, often through exit interviews and descriptive HR reports. Predictive analytics changes the timing of the question. Instead of asking only why employees left, an organization can examine patterns that appear before departures and use those patterns to improve working conditions, employee engagement, development opportunities, workload management, and retention planning. This project follows the CRISP DM framework described in the proposal: business understanding, data understanding, data preparation, modeling, evaluation, and business interpretation. The approach is intentionally end-to-end so that data quality, model performance, business usefulness, and ethical risk are considered together rather than treating predictive accuracy as the only goal.

The increasing availability of workforce data has created opportunities for organizations to take a more proactive approach to employee retention. Information about compensation, job satisfaction, workload, tenure, career development, and other employment characteristics can be analyzed together rather than examined separately. Machine learning is particularly useful in this setting because attrition may be associated with combinations of factors that are difficult to identify through basic reporting alone. However, the use of predictive analytics also increases the responsibility of organizations to distinguish statistical relationships from causal conclusions and to ensure that predictions are not used unfairly against employees.

Data Explanation

The analysis uses the IBM HR Analytics Employee Attrition and Performance dataset. The dataset is fictional and is intended for educational analysis rather than representation of actual IBM employees. The file contains 1,470 employee records and 35 original variables. The target variable, Attrition, identifies whether an employee stayed or left. The data include demographic information, job characteristics, compensation, satisfaction measures, work-life indicators, career history, and organizational variables. Initial quality checks found no missing values and no duplicate rows. Four fields with no useful predictive information or that function primarily as identifiers/constants - EmployeeCount, EmployeeNumber, Over18, and StandardHours - were removed from the modeling feature set. Attrition was encoded as 1 for Yes and 0 for No. Categorical predictors were one-hot encoded, numerical variables were standardized for logistic regression, and the data were divided into 80% training and 20% testing sets with stratification to preserve the attrition proportion.

This dataset includes a variety of numerical, categorical, and ordinal variables. Employee attrition can be viewed from a number of angles, such as from the perspective of an employee's compensation, or their career history. An employee can be viewed in terms of their numerical variables, such as their MonthlyIncome, and their Age, their TotalWorkingYears, and their YearsAtCompany. An employee can also be viewed in terms of their categorical variables, such as their JobRole, Department, BusinessTravel, and OverTime. An employee can also be viewed in terms of variables that determine their degree of satisfaction.

Next, we prepared the data to include only the relevant variables for our models. All identifiers and constant variables were removed from the dataset. Next, we performed one-hot encoding on the categorical variables in the classification models to include information from all of the categories in the models. We then split the data for training and testing purposes. We used a stratified split, as there are only a minority of employees who left the organization. Therefore, we needed to make sure that we preserved the proportion of employees who left the organization in both the training and testing datasets.

Methods

Three supervised classification methods were evaluated: Logistic Regression, Decision Tree, and Random Forest. Logistic Regression was used as the interpretable baseline because coefficients and feature effects can be communicated to nontechnical stakeholders. The Decision Tree provides understandable decision paths, while Random Forest tests whether an ensemble method improves classification. Because only a minority of employees in the dataset left the organization, class balanced weighting was used so the models did not simply optimize for the majority class. Performance was evaluated on a held-out test set using accuracy, precision, recall, F1 score, ROC-

AUC, and confusion matrices. The analysis places special emphasis on recall and ROC-AUC because failing to identify employees who leave can be more informative for a retention use case than maximizing overall accuracy alone.

Using three different classification algorithms made it possible to compare predictive performance with interpretability. Logistic Regression provides a relatively transparent baseline, while the Decision Tree offers understandable decision rules. Random Forest provides a more complex ensemble approach that can capture relationships that may not be represented as effectively by a simpler model. Comparing these approaches was important because the model with the highest accuracy is not automatically the most useful model for the business problem.

Particular attention was given to recall because the positive class represents employees who experienced attrition. A model with high accuracy but very low recall could appear successful while failing to identify most employees who actually left. ROC-AUC was also considered because it provides information about the model's ability to distinguish between the two classes across different classification thresholds. These measures provide a more complete evaluation than accuracy alone.

Analysis

The dataset contains 237 employees who left and 1233 who stayed, producing an overall attrition rate of 16.1%. This imbalance confirms why accuracy cannot be interpreted by itself.

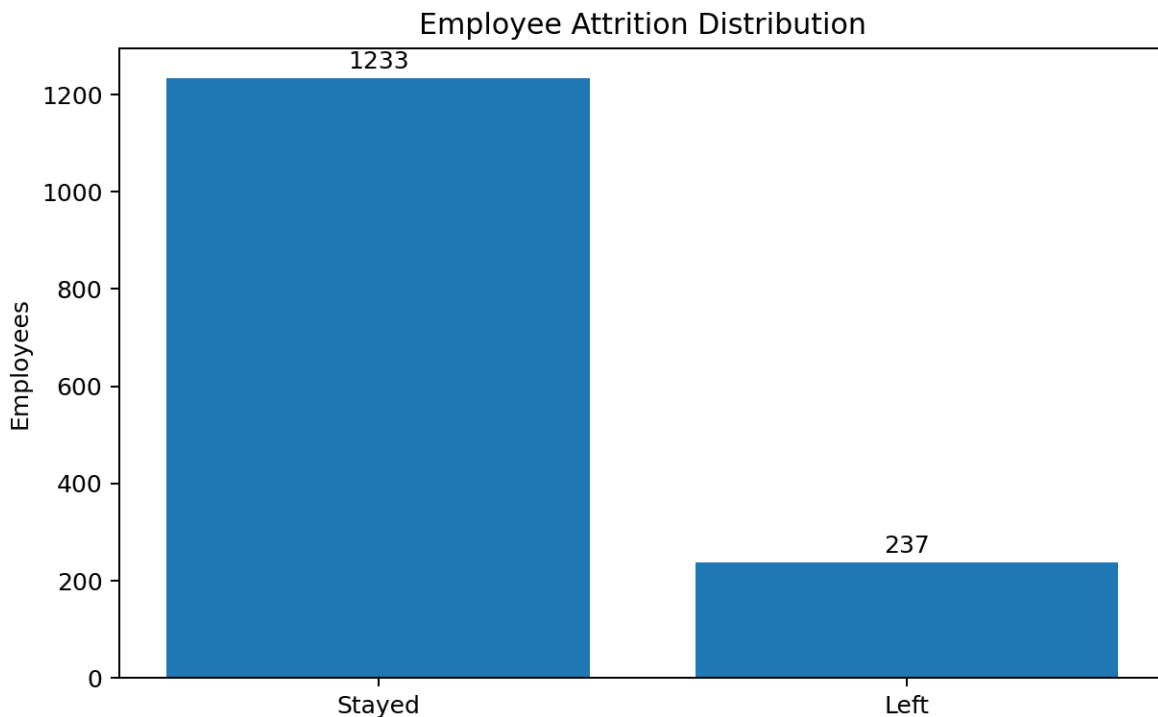


Figure 1. Attrition distribution in the IBM HR Analytics dataset.

Overtime showed one of the clearest differences in the exploratory analysis. Employees without overtime had an attrition rate of 10.4%, compared with 30.5% among employees who worked overtime. The observed rate for the overtime group was therefore almost three times the rate for

employees without overtime. This is an association rather than proof that overtime caused the departures, but it gives HR a practical area for further investigation.

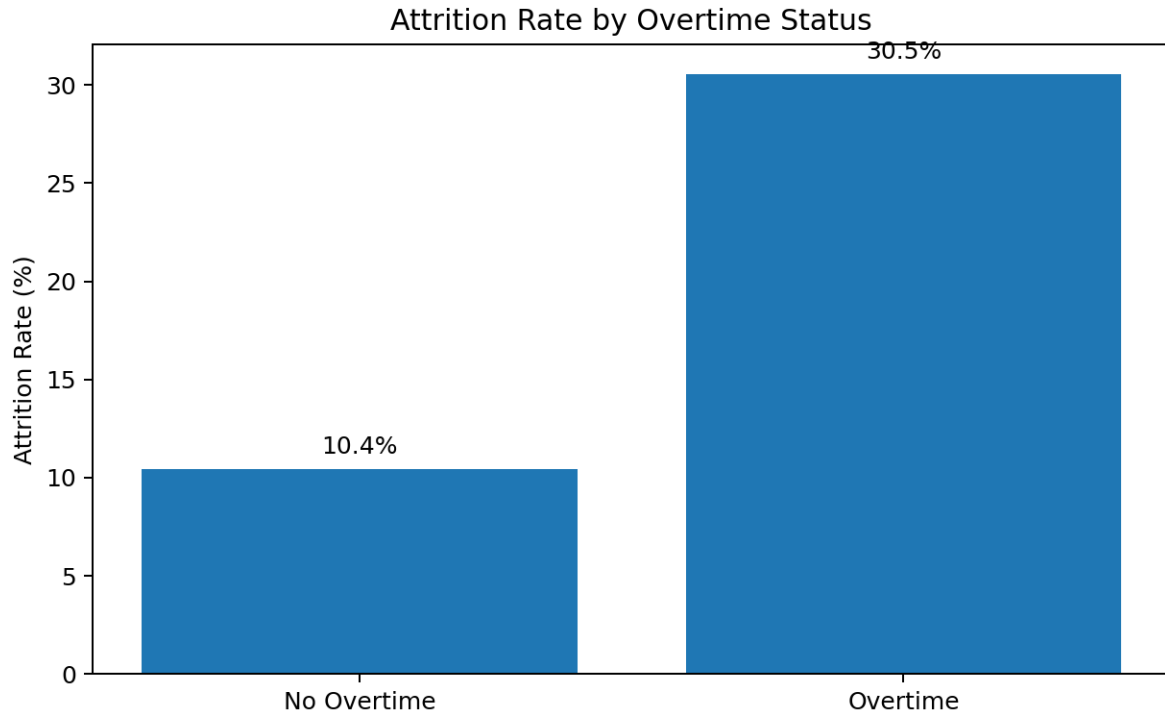


Figure 2. Attrition rate by overtime status. Overtime is associated with substantially higher observed attrition.

Compensation and tenure also differed between the two groups. Employees who stayed had an average monthly income of \$6,833, whereas employees who left averaged \$4,787. Median monthly income was \$5,204 for employees who stayed and \$3,202 for employees who left. Employees who stayed averaged 7.4 years at the company compared with 5.1 years among those who left. Job satisfaction showed a directional pattern as well: the attrition rate was 22.8% at the lowest job-satisfaction level and 11.3% at the highest level. Work-life balance at level 1 had the highest observed attrition rate at 31.2%.

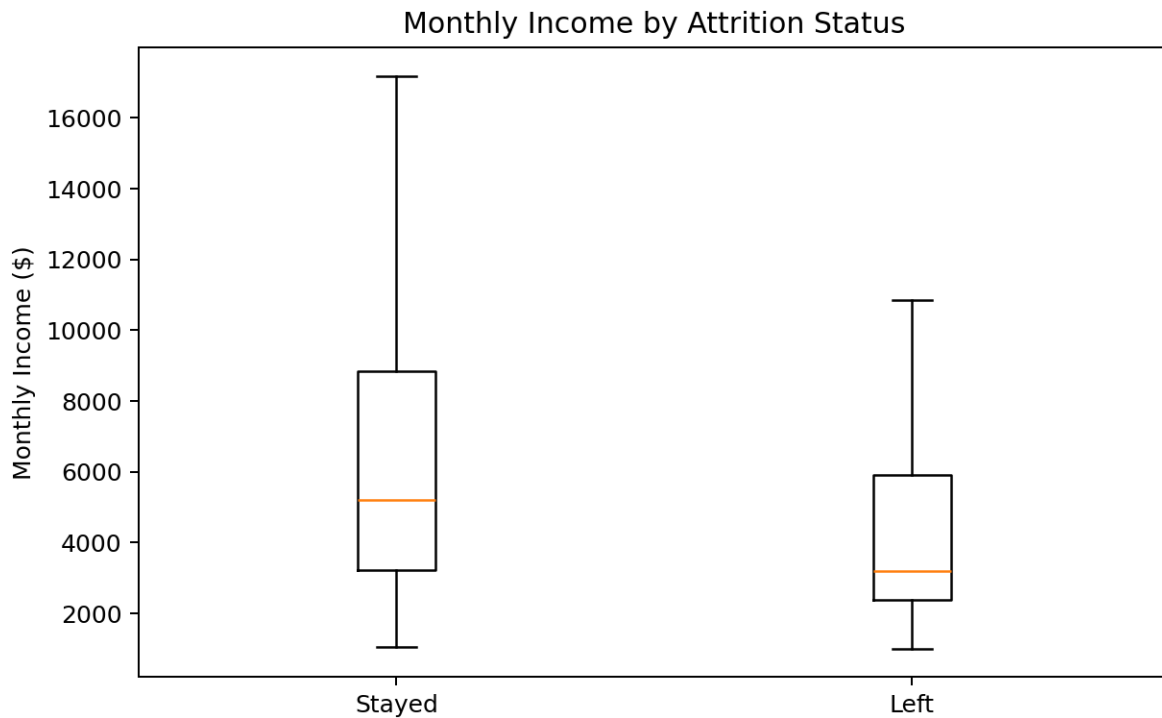


Figure 3. Monthly income distribution by attrition status.

The differences in compensation and tenure provide additional context for the attrition problem. Lower income among employees who left does not necessarily mean that compensation caused their departure because income is also related to factors such as job level, experience, and role. However, the finding provides a reason for organizations to examine whether compensation progression and career-development opportunities remain competitive for employees earlier in their careers.

The satisfaction results provide another important dimension. The attrition rate decreased from 22.8% at the lowest job-satisfaction level to 11.3% at the highest level, while employees reporting the lowest work-life balance experienced an attrition rate of 31.2%. Together with the overtime findings, these results suggest that employee retention should be examined as a combination of compensation, workload, career opportunities, and overall employee experience rather than as a problem with a single solution.

The models produced different tradeoffs. Random Forest achieved the highest overall accuracy at 0.833, but its recall for employees who left was only 0.149. Logistic Regression produced lower accuracy (0.752) but substantially stronger recall (0.638) and the highest ROC-AUC (0.803). The Decision Tree produced intermediate results, with 0.769 accuracy, 0.553 recall, and 0.702 ROC-AUC. For a retention decision-support use case, Logistic Regression is the most useful current candidate because it identifies a larger share of actual attrition cases and remains easier to explain to HR stakeholders.

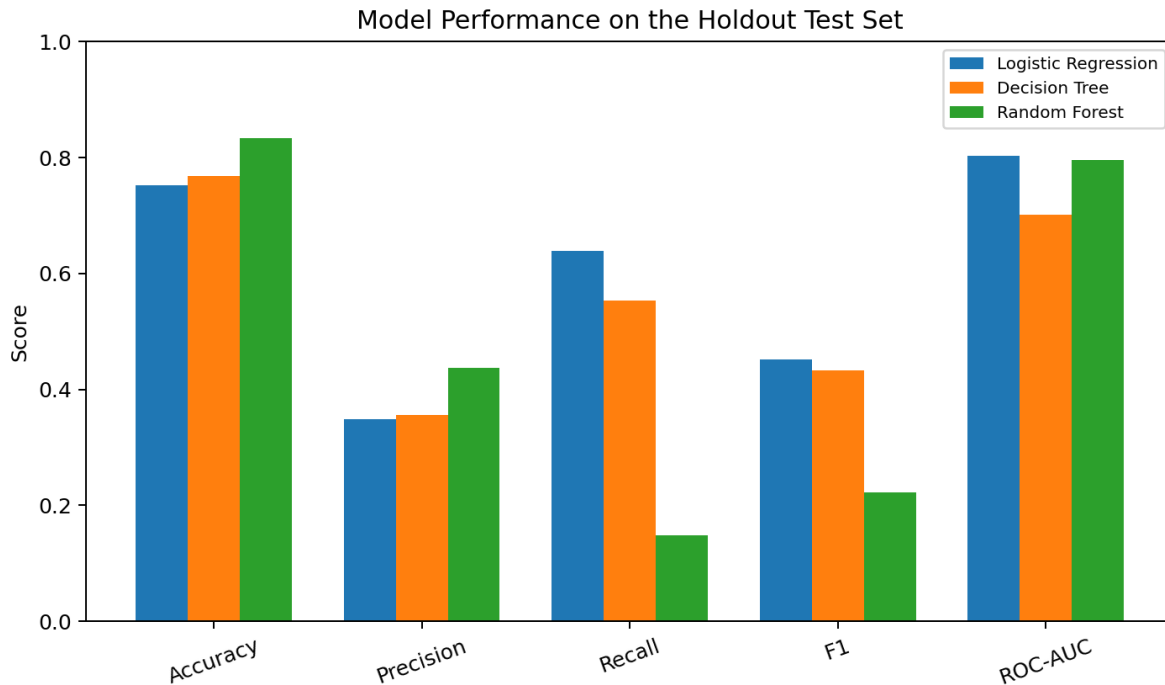


Figure 4. Comparison of classification metrics on the 20% holdout test set.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.752	0.349	0.638	0.451	0.803
Decision Tree	0.769	0.356	0.553	0.433	0.702
Random Forest	0.833	0.438	0.149	0.222	0.796

Permutation Importance for Logistic Regression pipeline. This plots the importance of the current model's variables in order of importance. The variables of interest are shown in the figure below. The variables are: overtime (strongest predictor), number of companies worked by, marital status, years since last promotion, job satisfaction, business travel, job role, job involvement, training frequency, years with the current manager.

The difference in magnitude between the two overtime groups would be of considerable value to any organization. In a hospital, for example, the HR department and the operational department would be able to see if there were departments, jobs, managers or times of the year of greater likelihood of overtime for certain employees. In particular, they would be able to find out if there were instances of employees working overtime due to under-staffing and therefore being forced to work overtime in order to complete their work. There could also be instances of poor scheduling and excessive workloads which are distributed unevenly amongst employees. In each of these instances, the causes of the instances of employees working overtime would need to be investigated and action taken in order to address the causes and prevent instances of employees being forced to work excessive hours in the future. Employees who work overtime are not necessarily intending to leave their employment.

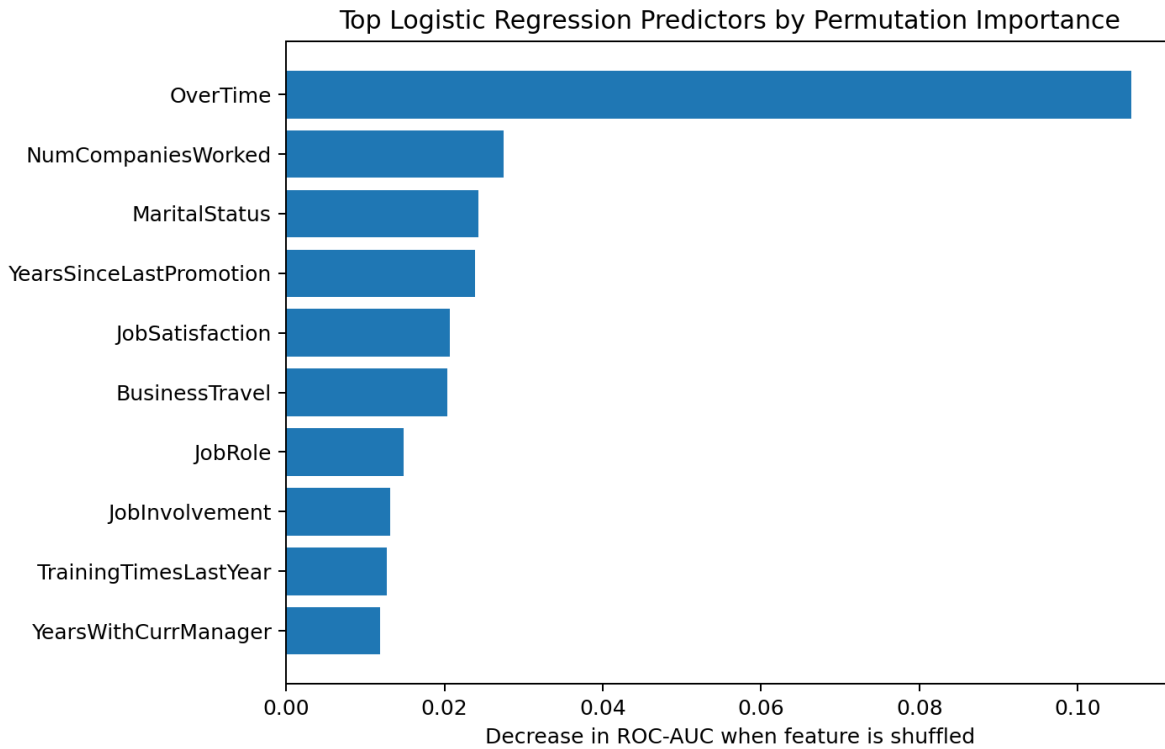


Figure 5. Top predictors from permutation importance for the Logistic Regression pipeline.

Conclusion

The draft analysis supports the original project objective: employee attrition contains measurable patterns that can be modeled, but the value of the model depends on how it is interpreted. Overtime, satisfaction, compensation, tenure, and career-history variables show meaningful differences between employees who stayed and those who left. Among the three tested models, Logistic Regression currently provides the strongest balance for HR decision support because its recall and ROC-AUC are stronger than the alternatives while its structure remains relatively transparent. The analysis does not justify labeling individual employees as likely to leave or using model output to make adverse employment decisions. Its strongest use is to identify organizational conditions that deserve attention.

Overall, the project demonstrates that the most valuable outcome of employee attrition modeling is not an individual risk score but a better understanding of organizational patterns. Overtime, compensation, satisfaction, tenure, and career history variables provide useful signals that can guide further investigation. The results also demonstrate that model selection must consider the purpose of the analysis. Although Random Forest achieved higher overall accuracy, Logistic Regression offered stronger recall and ROC-AUC while remaining easier to interpret.

The findings support a proactive approach to employee retention in which analytics is combined with organizational knowledge and human judgment. Rather than waiting until employees resign, organizations can examine workforce patterns and identify areas where changes in workload, development opportunities, compensation, management practices, or employee experience may be appropriate. Predictive analytics is therefore most valuable when it leads to better organizational questions and evidence-based improvements rather than automated decisions about individual employees.

Assumptions

This project assumes that the dataset definitions are internally consistent, that each row represents one employee, that Attrition is correctly coded, and that the supplied satisfaction and work-life scales can be treated as ordered indicators for exploratory interpretation. The train/test split assumes the records are independent observations. The analysis also assumes that the educational dataset is appropriate for demonstrating methodology even though it is not a real-time enterprise HR system.

The analysis further assumes that the relationships represented in the dataset are sufficiently meaningful for demonstrating classification methods. Satisfaction and work-life balance ratings are treated as ordered indicators, although the difference between two adjacent rating levels may not represent an equal change in employee experience. The analysis also assumes independence among employee observations. In a real organization, employees working for the same manager, department, or location may share workplace experiences, meaning that observations may not be completely independent.

Finally, the analysis assumes that predictive relationships are not equivalent to causal relationships. A variable may help a model distinguish between employees who stayed and employees who left without being the reason for the difference. This assumption is essential when translating model results into business recommendations.

Limitations

Limitation 1: The results from this study do not generalize to specific employers, industries, locations or workforces. This study was based off of a synthetic IBM dataset with 1,470 total observations. The number of observations in the attrition class was significantly less than the number of observations in the stay class. In some cases, the number of observations in the attrition class was less than the number of observations in a single split of the data. Therefore, the results from this study may be idiosyncratic to this specific dataset. Furthermore, the results from this study may not even generalize to the different splits of the data. In a final deployment study of this nature, one would use repeated cross-validation, perform calibration, check for fairness in different subgroups and lastly, use external data to validate the results from the study. Limitation 2: The study is cross-sectional in nature and does not enable the researcher to determine if causal relations that were found to hold in the data actually preceded the corresponding outcomes. Limitation 3: Several of the variables used in the study act as proxies for sensitive characteristics of employees or for organizations. These unobserved variables often have large effects on the results from a study of this nature. Limitation 4: This study only used one holdout split of the data.

Limitation 3: Even the best model does not capture all relevant variables and therefore only offers a partial explanation of the reasons for an employee's leave of absence. Other relevant reasons for an employee's leave of absence are for example the quality of leadership, the organization's culture, other employment opportunities outside the organization, an employee's family circumstances, economic conditions etc. Career goals which are changing can also lead to an employee's leave of absence.

Testing and validating a model before its application to a workforce is critical. This includes testing the model on current organizational data (. from the last few months, as well as from different departments or employee groups). External validation of the relationships found in this

research using real business data (. the educational data) is also crucial in order to verify whether these relationships hold true in a business environment.

Challenges

The main technical challenge was class imbalance. A model can obtain high accuracy by predicting that nearly everyone stays, which is not useful for retention planning. A second challenge is preprocessing the mixture of numerical and categorical variables without losing interpretability. A third challenge is selecting a model that performs reasonably well while remaining understandable to HR users. The largest nontechnical challenge is ethical: a statistically predictive feature may still be inappropriate for employee-level decision-making. The project therefore separates predictive usefulness from permission to use a variable operationally.

Future Uses/Additional Applications

With real organizational data and appropriate governance, the same workflow could be extended to department-level retention monitoring, workload analysis, promotion and career-path evaluation, onboarding risk analysis, training effectiveness, engagement surveys, and workforce planning. Future work could also evaluate model drift over time, compare outcomes across locations or job families, and add explainability tools such as SHAP values. These extensions should focus on organizational improvement and resource planning rather than surveillance of individual employees.

Recommendations

The first recommendation is to treat overtime as an organizational signal worthy of review because the observed attrition rate is substantially higher among employees who work overtime. HR and operations leaders should investigate whether overtime reflects chronic understaffing, uneven workload distribution, scheduling pressure, or role-specific demands. Second, organizations should review retention conditions for lower-income and shorter-tenure employees and pair compensation analysis with job-satisfaction and development data. Third, the model should be used at aggregated team or workforce levels whenever possible. Sensitive or potentially discriminatory variables should be excluded from operational decision rules unless there is a clearly justified, lawful, and fairness-tested purpose. Finally, model success should be measured not only by predictive metrics but also by whether interventions improve employee experience without creating unfair treatment.

Implementation Plan

Responsible implementation starts with a pilot. Step 1: Validate workflow with currently governed HR data and list all fields that are being used. Step 2: Remove all unnecessary identifiers and decide on protected variables and/or proxy variables in business requirements meetings to exclude from training as well as from scoring in operational context. Step 3: Train and validate the most simple, yet interpretable baseline models and compare results with the same metrics to corresponding ensemble methods (e.g. Random Forest, Gradient Boost). Review recall, precision, ROC-AUC, calibration as well as sub-group performance. Step 4: Present aggregated findings from previous steps to appropriate HR leaders and subject matter experts for contextual review. Step 5: Design retention interventions targeting the following workplace conditions instead of individual employees: workload, manager support, employee development opportunities and fair scheduling practices. Step 6: Continuously monitor outcomes, model drift, false positives, false negatives and fairness metrics for all employment-related decisions that are being made with the aid of this

automated system. Also establish a documented human review process for decisions made with the aid of this automated system.

Ethical Assessment

Predictive HR analytics creates significant fairness, privacy, transparency, and governance risks. Historical workforce data may encode past inequities, and a model can reproduce those patterns even when it appears mathematically accurate. Employee data can include sensitive demographic, compensation, performance, and career information, so access should be restricted and identifiers minimized. Model predictions should never automatically determine hiring, promotion, compensation, discipline, or termination. The model should instead help identify workplace conditions that management can improve.

Fairness testing is especially important because the current feature importance analysis includes variables such as marital status alongside job-related factors. A production system should test performance across relevant employee groups, document disparate error rates, and consider removing variables that introduce unacceptable risk or are not necessary for the business purpose. Transparency is also essential: stakeholders should understand that feature importance describes association, not causation, and that a high predicted probability does not establish an employee's intent to resign. These safeguards directly address the central ethical concern raised in the Milestone 1 feedback: employee-level predictions can have serious consequences and must remain subordinate to human judgment, privacy protections, and organizational accountability.

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Appendix A: Supporting Documentation

Selected Data Dictionary

Variable	Description
Attrition	Target variable: Yes = employee left; No = employee stayed.
Age	Employee age in years.
BusinessTravel	Frequency of business travel.
Department	Organizational department.
DistanceFromHome	Distance from home to work.
EnvironmentSatisfaction	Ordinal satisfaction rating from 1 (low) to 4 (very high).
JobInvolvement	Ordinal involvement rating from 1 to 4.
JobLevel	Job level within the organization.
JobRole	Employee job role.
JobSatisfaction	Ordinal satisfaction rating from 1 to 4.
MonthlyIncome	Monthly income amount.
OverTime	Whether the employee works overtime.
TotalWorkingYears	Total years of work experience.
WorkLifeBalance	Ordinal balance rating from 1 to 4.
YearsAtCompany	Years employed at the current organization.
YearsSinceLastPromotion	Years since the employee was last promoted.
YearsWithCurrManager	Years working with the current manager.

Modeling Configuration

- Target: Attrition (Yes = 1, No = 0)
- Train/test split: 80/20 with stratification and random_state = 42
- Categorical preprocessing: one-hot encoding
- Numerical preprocessing: standardization for Logistic Regression
- Models: Logistic Regression, Decision Tree, Random Forest
- Class handling: balanced class weights
- Primary evaluation metrics: recall, F1, ROC-AUC, precision, accuracy, and confusion matrix

Ten Anticipated Audience Questions

1. Why was Logistic Regression selected as the preferred draft model when Random Forest had higher accuracy?
2. How does class imbalance affect the interpretation of accuracy in this dataset?
3. Does the higher attrition rate among overtime employees prove that overtime causes employees to leave?
4. Which variables had the strongest influence on the model, and how were they measured?

5. Why should marital status or other sensitive personal variables be treated cautiously even if they improve prediction?
6. How would the model be validated before using it with real organizational data?
7. What should HR do when an employee or group is identified as having elevated attrition risk?
8. How can the organization protect employee privacy when performing predictive workforce analytics?
9. What types of bias or fairness checks should be completed before deployment?
10. How would the model be monitored over time to determine whether its predictions remain reliable?

Reproducibility Note

The analysis was completed in Python using pandas and scikit-learn. The source dataset contains 1,470 rows and 35 original columns. All descriptive statistics, model metrics, and figures in this draft were recalculated directly from the dataset used for this Milestone 2 draft. Random-state values were fixed where applicable to support reproducibility.